

12/6/96

Week-2 Module 1

1. Find Student Roll No. (Linear Search)

```
import java.util.Scanner;
public class demo {
    public static void main (String[]
        args) {
        Scanner sc = new Scanner (System.in);
        int n = sc.nextInt();
        int [] rollNumbers = new int[n];
        for (int i=0; i<n; i++) {
            rollNumbers[i] = sc.nextInt();
        }

        int x = sc.nextInt();
        int foundIndex = -1;
        for (int i=0; i<n; i++) {
            if (rollNumbers[i] == x) {
                foundIndex = i;
                break;
            }
        }
    }
}
```

```

if (foundIndex != -1) {
    System.out.println("Found at
                        Index" + foundIndex);
}
else {
    System.out.println("Not Found");
}
sc.close();
}
}

```

2. Search Product ID (Binary Search)

```

import java.util.Scanner;
public class demo {
    public static boolean binarySearch
        (int[] products, int target) {
        int left = 0;
        int right = products.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (products[mid] == target) {

```



```

return true;
}
if (products[mid] < target) {
    left = mid + 1;
}
else {
    right = mid - 1;
}
}
return false;
}

```

```

public static void main (String[] args) {
    Scanner sc = new Scanner (System.in);
    if (sc.hasNextInt ()) {

```

```

        int n = scanner.nextInt();
        int[] productID = new int [n];
        for (int i = 0; i < n; i++) {
            productID[i] = scanner.nextInt();
        }

```

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```

        int targetID = scanner.nextInt();
        if (binarySearch (productID,
            targetID)) {

```

```
System.out.println("Product  
Available");
```

```
}
```

```
else {
```

```
System.out.println("Product  
not Available");
```

```
}
```

```
}
```

```
scanner.close();
```

```
}
```

```
}
```

3. Arrange exam scores (Bubble sort)

```
import java.util.Scanner;
```

```
public class demo {
```

```
public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

```
if (sc.hasNextInt()) {
```

```
int n = sc.nextInt();
```

```
int[] scores = new int[n];
```

```
for (int i = 0; i < n; i++) {
```

```
scores[i] = sc.nextInt();
```

```
}
```



```

for (int i=0; i<n-1; i++) {
    for (int j=0; j<n-i-1; j++) {
        if (scores[j] > scores[j+1]) {
            int temp = scores[j];
            scores[j] = scores[j+1];
            scores[j+1] = temp;
        }
    }
}

for (int i=0; i<n; i++) {
    System.out.print(score[i]+" ");
}

sc.close();
}
}

```

Terminal

5

12 48 52 3 28

3 12 28 48 52

4. Arrange Library Book No's. (Insertion Sort)

```
import java.util.Scanner;
public class demo {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        if (sc.hasNextInt ()) {
            int n = sc.nextInt ();
            int [] books = new int [n];
            for (int i = 0; i < n; i++) {
                books [i] = sc.nextInt ();
            }
            for (int i = 1; i < n; i++) {
                int key = books [i];
                int j = i - 1;
                while (j >= 0 & books [j] > key) {
                    books [j+1] = books [j];
                    j--;
                }
                books [j+1] = key;
            }
            for (int i = 0; i < n; i++) {
                System.out.print (books [i] + " ");
            }
        }
    }
}
```



```

    }
    sc.close();
}

```

Terminal

5

```

55 23 87 11 34
11 23 34 55 87

```

5: Rank Players by Score (selection sort)

```

import java.util.Scanner;
public class demo {
    public static void main (String[] args) {
Scanner sc = sc.next();
        Scanner sc = new Scanner (System.in);
        if (sc.hasNextInt()) {
            int n = sc.nextInt();
            int [] scores = new int [n];
            for (int i=0; i<n; i++) {
                scores [i] = sc.nextInt();
            }

```

```

for (int i = 0; i < n - 1; i++) {
    int minIdx = i;
    for (int j = i + 1; j < n; j++) {
        if (scores[j] < scores[minIdx]) {
            minIdx = j;
        }
    }
    int temp = scores[minIdx];
    scores[minIdx] = scores[i];
    scores[i] = temp;
}
for (int i = 0; i < n; i++) {
    System.out.print(scores[i] + " ");
}
}
sc.close();
}
}

```

Terminal

```

5
70 40 90 10 20
10 20 40 70 90

```


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emo.class

emo.java

v.json

J demo.java X

.vscode > r coding > J demo.java > demo > main(String[])

```

1  import java.util.Scanner;
2
3  public class demo {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          int n = sc.nextInt();
8          int[] rollNumbers = new int[n];
9          for (int i = 0; i < n; i++) {
10             rollNumbers[i] = sc.nextInt();
11         }
12         int x = sc.nextInt();
13         int foundIndex = -1;
14         for (int i = 0; i < n; i++) {
15             if (rollNumbers[i] == x) {
16                 foundIndex = i;
17                 break;
18             }
19         }
20         if (foundIndex != -1) {
21             System.out.println("Found at index " + foundIndex);
22         } else {
23             System.out.println(x: "Not Found");
24         }
25         sc.close();
26     }
27 }

```

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ROJECTS

vscode

r coding

demo +

JRE System Lib...

Referenced Lib...

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```

PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
101 105 103 110 112
110
Found at index 3
PS C:\Users\ym\.vscode\r coding>

```

J demo.java X

.vscode > r coding > J demo.java > demo > main(String[])

```
1  import java.util.Scanner;
2
3  public class demo {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          int n = sc.nextInt();
8          int[] rollNumbers = new int[n];
9          for (int i = 0; i < n; i++) {
10             rollNumbers[i] = sc.nextInt();
11         }
12         int x = sc.nextInt();
13         int foundIndex = -1;
14         for (int i = 0; i < n; i++) {
15             if (rollNumbers[i] == x) {
16                 foundIndex = i;
17                 break;
18             }
19         }
20         if (foundIndex != -1) {
21             System.out.println("Found at index " + foundIndex);
22         } else {
23             System.out.println(x: "Not Found");
24         }
25         sc.close();
26     }
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
101 105 103 110 112
110
Found at index 3
PS C:\Users\ym\.vscode\r coding> & 'c:\Users\ym\.vscode\r coding\demo.java'
PS C:\Users\ym\.vscode\r coding> 5
5
PS C:\Users\ym\.vscode\r coding> & 'c:\Users\ym\.vscode\r coding\demo.java'
PS C:\Users\ym\.vscode\r coding> ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
101 105 113 110 130
150
Not Found
PS C:\Users\ym\.vscode\r coding> |
```


demo.java X

```
.vscode > r coding > demo.java > demo > main(String[])
```

```
1  import java.util.Scanner;
2
3  public class demo {
4      public static boolean binarySearch(int[] products, int target) {
5          int left = 0;
6          int right = products.length - 1;
7
8          while (left <= right) {
9              int mid = left + (right - left) / 2;
10             if (products[mid] == target) {
11                 return true;
12             }
13             if (products[mid] < target) {
14                 left = mid + 1;
15             }
16             else {
17                 right = mid - 1;
18             }
19         }
20         return false;
21     }
22
23     Run | Debug
24     public static void main(String[] args) {
25         Scanner scanner = new Scanner(System.in);
26         if (scanner.hasNextInt()) {
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
10 20 30 40 50
40
Product Available
PS C:\Users\ym\.vscode\r coding>
```

J demo.java X

.vscode > r coding > J demo.java > demo > main(String[])

```
1  import java.util.Scanner;
2
3  public class demo {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          if (sc.hasNextInt()) {
8              int n = sc.nextInt();
9              int[] scores = new int[n];
10             for (int i = 0; i < n; i++) {
11                 scores[i] = sc.nextInt();
12             }
13             for (int i = 0; i < n - 1; i++) {
14                 for (int j = 0; j < n - i - 1; j++) {
15                     if (scores[j] > scores[j + 1]) {
16                         int temp = scores[j];
17                         scores[j] = scores[j + 1];
18                         scores[j + 1] = temp;
19                     }
20                 }
21             }
22             for (int i = 0; i < n; i++) {
23                 System.out.print(scores[i] + " ");
24             }
25             sc.close();
26         }
27     }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
12 48 52 3 28
3 12 28 48 52
PS C:\Users\ym\.vscode\r coding> 
```


J demo.java X

.vscode > r coding > J demo.java > demo > main(String[])

```
1  import java.util.Scanner;
2  public class demo {
3      Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          if (sc.hasNextInt()) {
7              int n = sc.nextInt();
8              int[] books = new int[n];
9              for (int i = 0; i < n; i++) {
10                 books[i] = sc.nextInt();
11             }
12             for (int i = 1; i < n; i++) {
13                 int key = books[i];
14                 int j = i - 1;
15                 while (j >= 0 && books[j] > key) {
16                     books[j + 1] = books[j];
17                     j--;
18                 }
19                 books[j + 1] = key;
20             }
21             for (int i = 0; i < n; i++) {
22                 System.out.print(books[i] + " ");
23             }
24             sc.close();
25         }
26     }
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ym\.vscode> cd "c:\Users\ym\.vscode\r coding\" ; if ($?) { javac demo.java } ; if ($?) { java demo }
5
55 23 87 11 34
11 23 34 55 87
PS C:\Users\ym\.vscode\r coding>
```

J demo.java X

.vscode > r coding > J demo.java > demo > main(String[])

```
1  import java.util.Scanner;
2  public class demo {
    Run | Debug
3      public static void main(String[] args) {
4          Scanner sc = new Scanner(System.in);
5          if (sc.hasNextInt()) {
6              int n = sc.nextInt();
7              int[] scores = new int[n];
8              for (int i = 0; i < n; i++) {
9                  scores[i] = sc.nextInt();
10             }
11             for (int i = 0; i < n - 1; i++) {
12                 int minIdx = i;
13                 for (int j = i + 1; j < n; j++) {
14                     if (scores[j] < scores[minIdx]) {
15                         minIdx = j;
16                     }
17                 }
18                 int temp = scores[minIdx];
19                 scores[minIdx] = scores[i];
20                 scores[i] = temp;
21             }
22             for (int i = 0; i < n; i++) {
23                 System.out.print(scores[i] + " ");
24             }
25         }
26         sc.close();
27     }
28 }
```

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```
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5
70 40 90 10 20
10 20 40 70 90
PS C:\Users\ym\.vscode\r coding>
```